

Question	Answer	Mark	Mark scheme	Additional guidance
20 (a)	Shown	M1	$\frac{1}{9} \times \frac{1}{8} = \frac{1}{72}$ for $\frac{1}{9} \times \frac{1}{8} (= \frac{1}{72})$ or $9 \times 8 (= 72)$ C1 for a complete argument $\frac{1}{72} + \frac{1}{72} = \frac{1}{36}$ eg	
(b)	$\frac{8}{84}$	P1	$\frac{1}{9} \times \frac{1}{8} \times \frac{1}{7} = \frac{1}{504}$ for $\frac{1}{9} \times \frac{1}{8} \times \frac{1}{7} (= \frac{1}{504})$ OR finds that there are 8 possible lines of 3	
		P1	for process to find probability of one particular arrangement of 3 circles $\frac{1}{9} \times \frac{1}{8} \times \frac{1}{7} \times 6 \text{ oe } (= \frac{1}{84})$ eg $\frac{1}{9} \times \frac{1}{8} \times \frac{1}{7} \times 6 \text{ oe } (= \frac{1}{84})$ OR process to find number of different arrangements of 3 circles eg $9 \times 8 \times 7 \div 6 (= 84)$	
		A1	$\frac{8}{84}$ for $\frac{8}{84}$ oe	